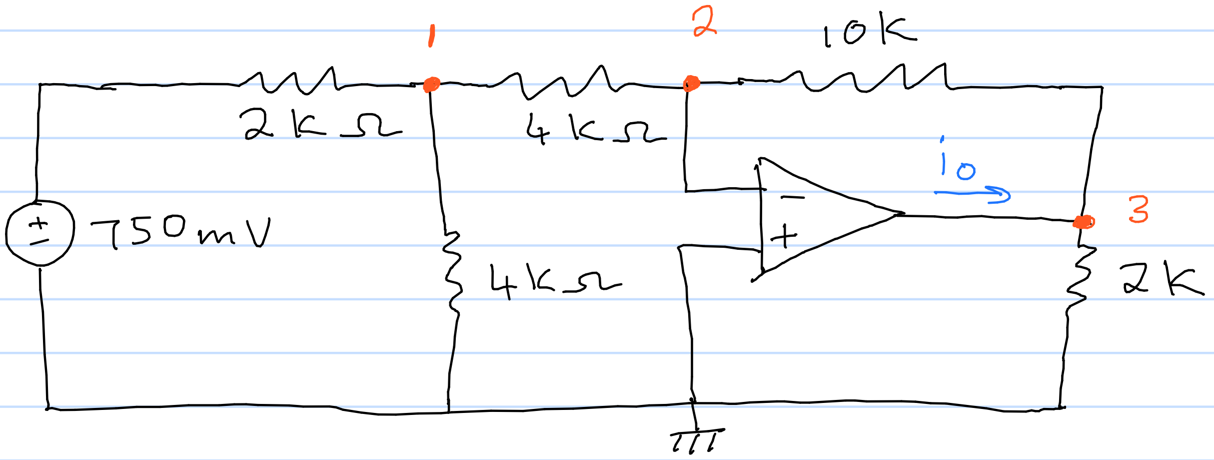
5.19 Determine i_o



At node 1: KCL

$$\frac{0{,}75 - V_1}{2000} = \frac{V_1}{4000} + \frac{V_1}{4000}$$

$$\frac{0{,}75 - V_1}{2000} = \frac{V_1}{2000}$$

$$0{,}75 = 2V_1$$

$$V_1 = 0{,}375$$

At node 2: KCL

$$\frac{0{,}375}{4000} = \frac{-V_o}{10\,000}$$

$$V_o = \frac{-0{,}375 \times 10000}{4000}$$

$$V_o = -0{,}9375$$

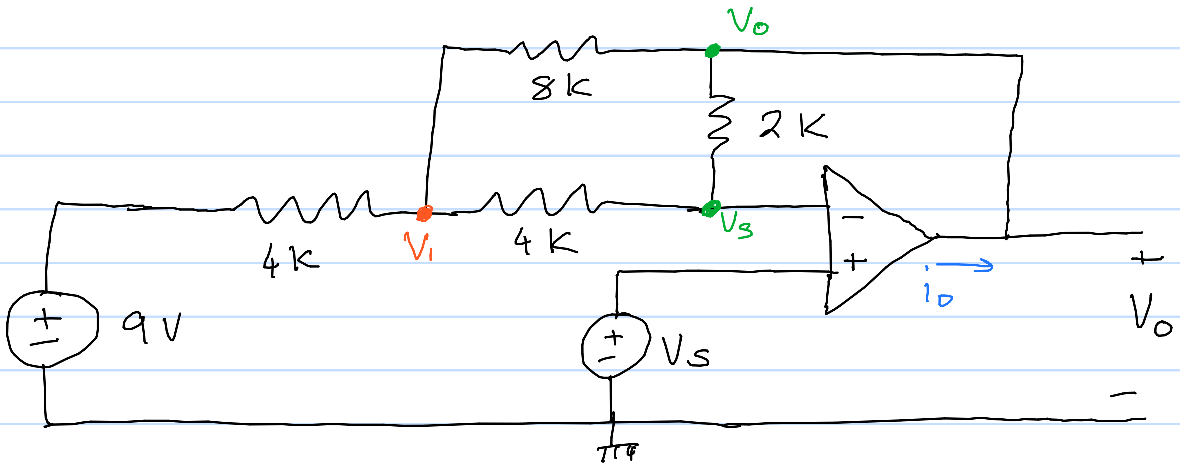
At node 3: KCL

$$\frac{0{,}375 - (-0{,}9375)}{14000} + i_o = \frac{-0{,}9375}{2000}$$

$$i_o = -468{,}75\,\mu A - 93{,}75\,\mu A$$

$$i_o = -562{,}5\,\mu A$$

5.20 Calculate V_o if $V_s = 2\,V$



KCL at V_1:

$$\frac{9 - V_1}{4000} + \frac{V_s - V_1}{4000} + \frac{V_o - V_1}{8000} = 0 \qquad (1)$$

KCL at V_o:

$$\frac{V_1 - V_o}{8000} + \frac{V_s - V_o}{2000} + i_o = 0 \qquad (2)$$

KCL at V_s:

$$\frac{V_1 - V_s}{4000} + \frac{V_o - V_s}{2000} = 0 \qquad (3)$$

* Equation 1 & 3 are sufficient to find V_o.

$$18 - 2V_1 + 2V_s - 2V_1 + V_o - V_1 = 0 \qquad (1)$$

$$18 - 2V_1 + 2(2) - 2V_1 + V_o - V_1 = 0$$

$$V_o - 5V_1 = -22$$

$$V_1 - V_s + 2V_o - 2V_s = 0$$

$$V_1 - 2 + 2V_o - 2(2) = 0$$

$$2V_o + V_1 = 6$$

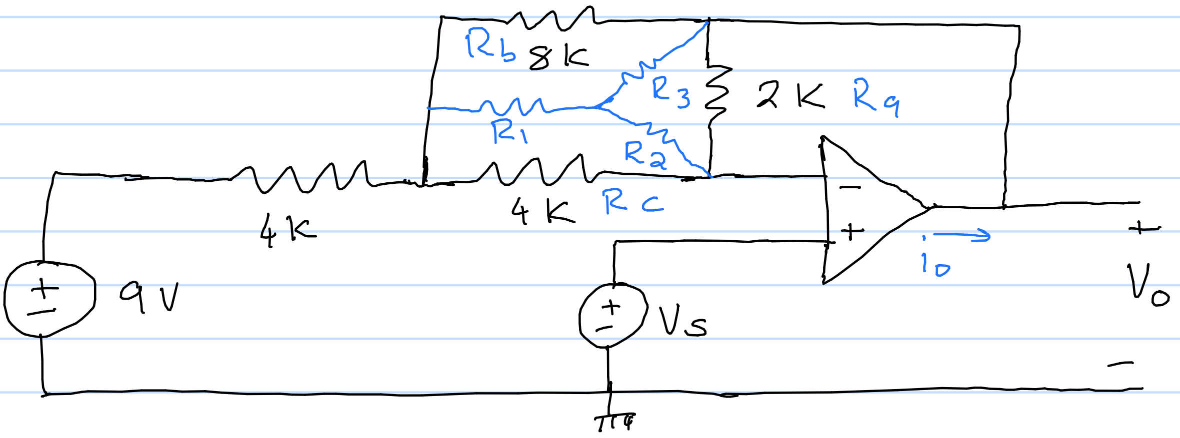
Set up and solve matrix equation

$$\begin{bmatrix} 1 & -5 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} V_o \\ V_1 \end{bmatrix} = \begin{bmatrix} -22 \\ 6 \end{bmatrix}$$

$V_o = 0.727273 = 727.273\,mV$

$V_1 = 4.545\ V$

5.20 Calculate V_o if $V_s = 2\,V$



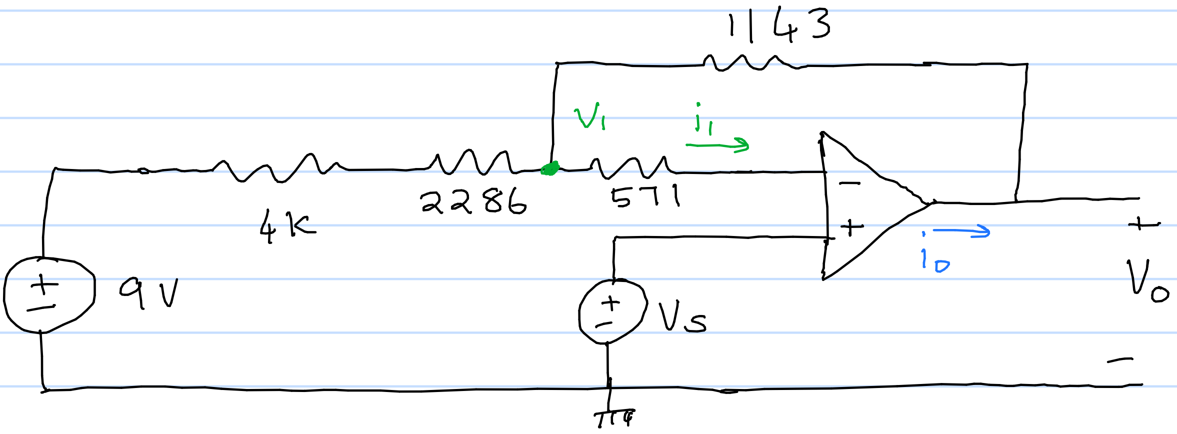
$\Delta - Y$ transform

$$R_a + R_b + R_c = 2000 + 8000 + 4000$$
$$= 14\,000$$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{8000 \times 4000}{14\,000} = 2286\,\Omega$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} = \frac{4000 \times 2000}{14000} = 571\,\Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{2000 \times 8000}{14\,000} = 1143\,\Omega$$



Since $i_1 = 0$, $V_1 = V^- = V_s$

$$\frac{9 - V_s}{6286} = \frac{V_s - V_o}{1143}$$

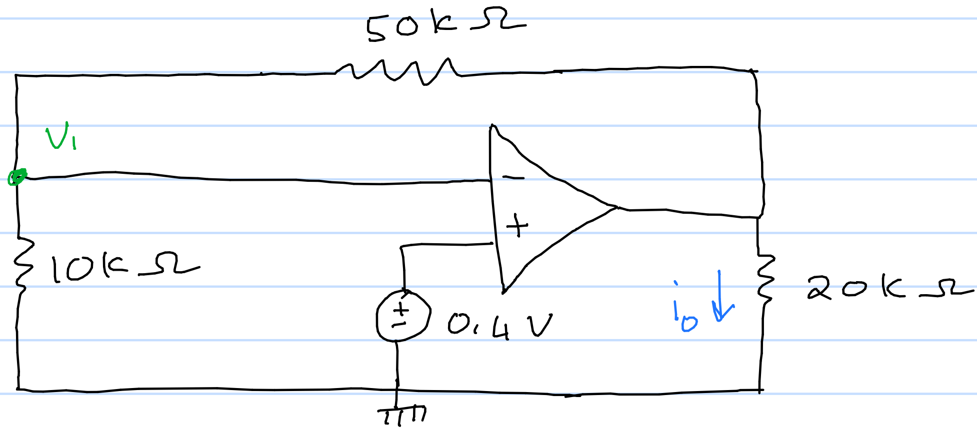
$$\frac{9-2}{6286} = \frac{2}{1143} - \frac{V_o}{1143}$$

$$V_o = \left(\frac{7}{6286} - \frac{2}{1143}\right)(-1143)$$

$$V_o = 0{,}727\ V$$

$$= 727{,}171\ mV$$

5.28 Find i_o



KCL at node V_1:

$$\frac{-V_1}{10000} = \frac{V_1 - V_o}{50\,000}$$

$$\frac{-0.4}{10000} = \frac{0.4 - V_o}{50000}$$

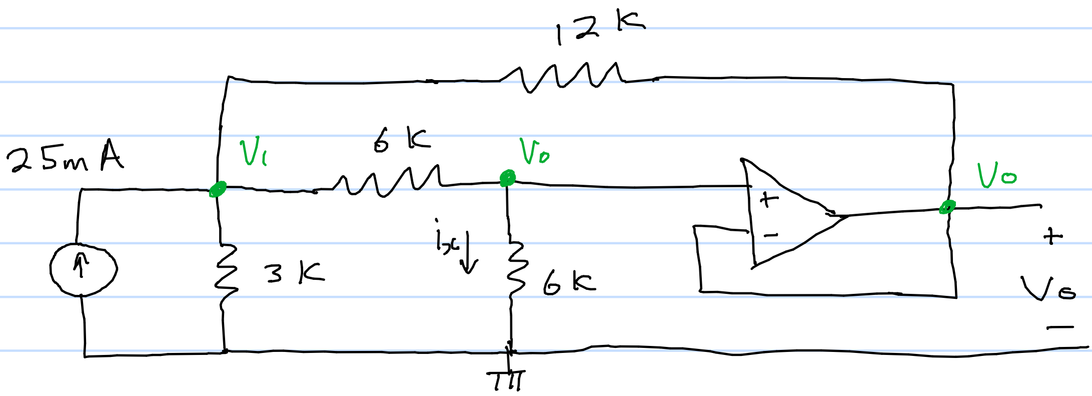
$$V_o = \left(\frac{-0.4}{10000} - \frac{0.4}{50000}\right)(-50000)$$

$$V_o = 2.4\ V$$

$$i_o = \frac{V_o}{20\,000}$$

$$i_o = 120\,\mu A$$

5.31 find i_x



KCL at V_1

$$0.025 = \frac{V_1}{3000} + \frac{V_1 - V_o}{6000} + \frac{V_1 - V_o}{12000}$$

$$300 = 4V_1 + 2V_1 - 2V_o + V_1 - V_o$$

$$300 = -3V_o + 7V_1$$

KCL at V_o

$$\frac{V_1 - V_o}{6000} + \frac{0 - V_o}{6000} = 0$$

$$V_1 - V_o \quad - V_o = 0$$

$$-2V_o + V_1 = 0$$

Set up and solve matrix equation

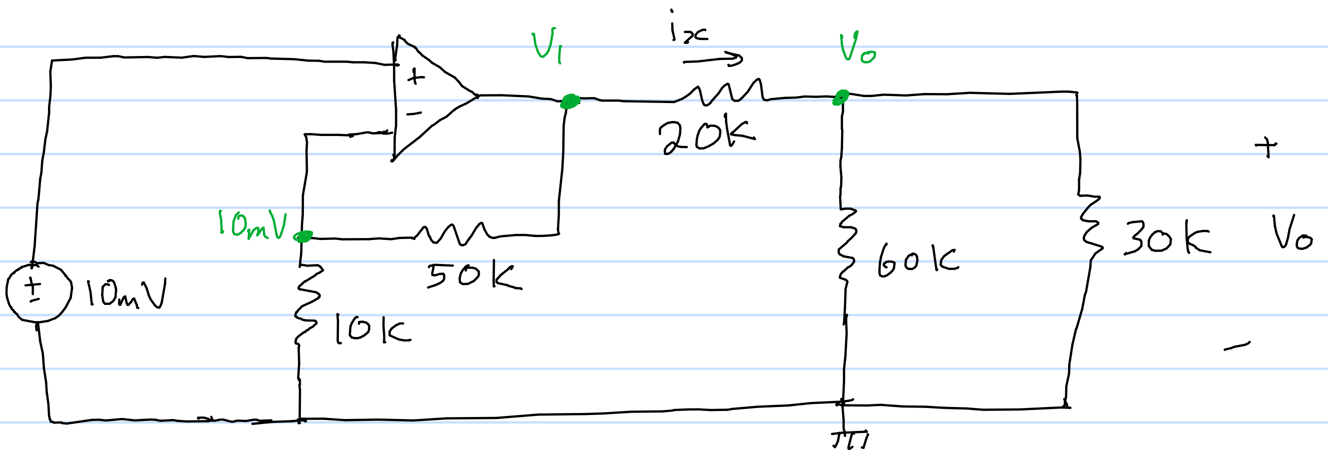
$$\begin{bmatrix} -3 & 7 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} V_o \\ V_1 \end{bmatrix} = \begin{bmatrix} 300 \\ 0 \end{bmatrix}$$

$$V_o = 27.273 \text{ V}$$

$$V_1 = 54.545 \text{ V}$$

$$i_x = \frac{V_o}{6000} = \frac{27.273}{6000} = 4.55 \text{ mA}$$

5.32 Calculate i_x, V_o and the power dissipated in the $60k\Omega$ resistor



KCL at 10mV node:

$$\frac{V_1 - 0.01}{5000} = \frac{0.01}{10000}$$

$$V_1 = \left(\frac{0.01}{10000} + \frac{0.01}{50000}\right) 50\,000$$

$$V_1 = 0.06$$

Find V_o through voltage division

$$V_o = \frac{V_1 \times 60\,000 \| 30\,000}{60\,000 \| 30\,000 + 20000}$$

$$V_o = 0.03 = 30\ mV$$

$$i_x = \frac{V_1 - V_o}{20\,000}$$

$$= \frac{0.06 - 0.03}{20\,000}$$

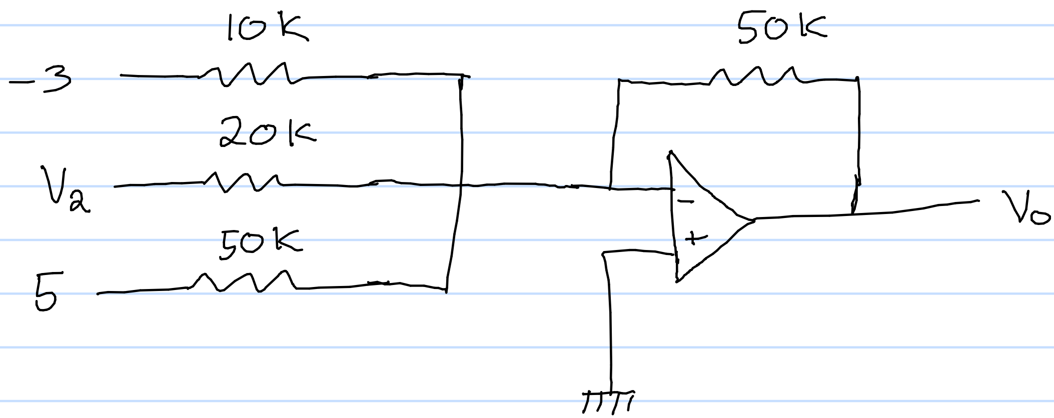
$$= 1.5\ \mu A$$

$$P_{60K} = \frac{V_o^2}{60\,000}$$

$$= \frac{0.03^2}{60\,000}$$

$$= 15\ nW$$

5,39 find V_2 such that $V_o = -16.5$ V



Summing amplifier

$$V_o = -\left(\frac{50000}{10000} \times (-3) + \frac{50000}{20000} V_2 + \frac{50000}{50000} \times 5\right)$$

$$-16.5 = 15 - \frac{5}{2} V_2 - 5$$

$$V_2 = (-16.5 - 15 + 5) \times \frac{-2}{5}$$

$$V_2 = 10.6 \text{ V}$$

5.45 Design an op-amp circuit to perfom the following operation:

$$V_o = 3.5 V_1 - 2.5 V_2$$

* Difference amplifier

$$V_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} V_2 - \frac{R_2}{R_1} V_1$$

$$\frac{R_2}{R_1} = 2.5$$

$$\frac{2.5(1 + 2.5^{-1})}{1 + R_3/R_4} = 3.5$$

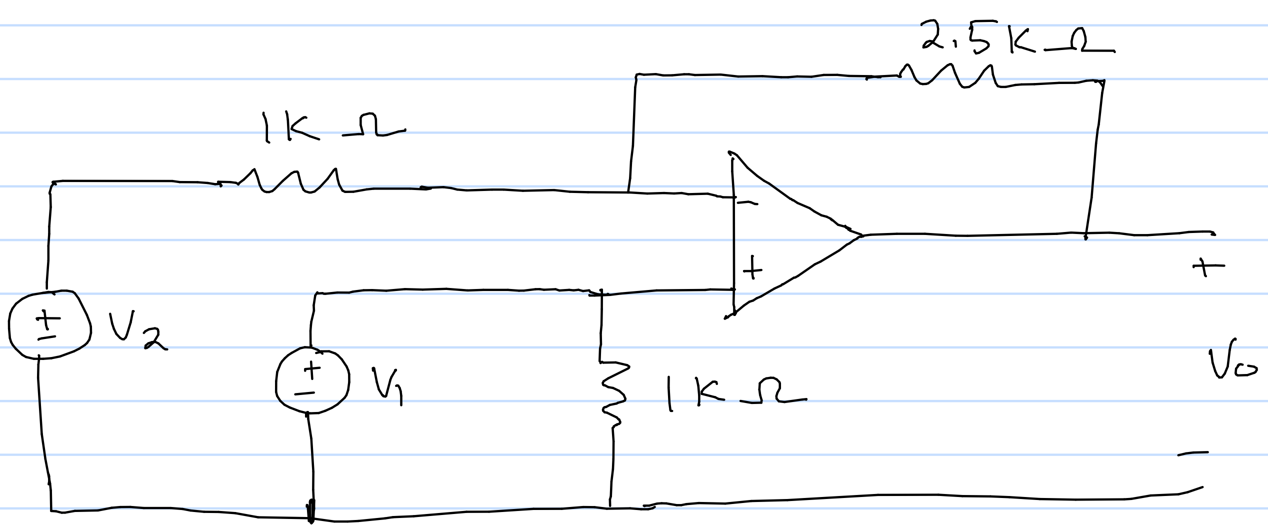
$$2.5(1.4) = 3.5 + \frac{R_3}{R_4} 3.5$$

$$3.5 - 3.5 = \frac{R_3}{R_4} 3.5$$

$$\frac{R_3}{R_4} = 0$$

$R_1 = 1\,K\Omega$ $R_2 = 2.5\,K\Omega$

$R_3 = 0\,\Omega$ $R_4 = 1\,K\Omega$



5.45 Design an op-amp circuit to perfom the following operation:

$$V_o = 3.5 V_1 - 2.5 V_2$$

* Summing amplifier

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2\right)$$

* Inverting amplifier

$$V_o = -\frac{R_f}{R_1} V_i$$

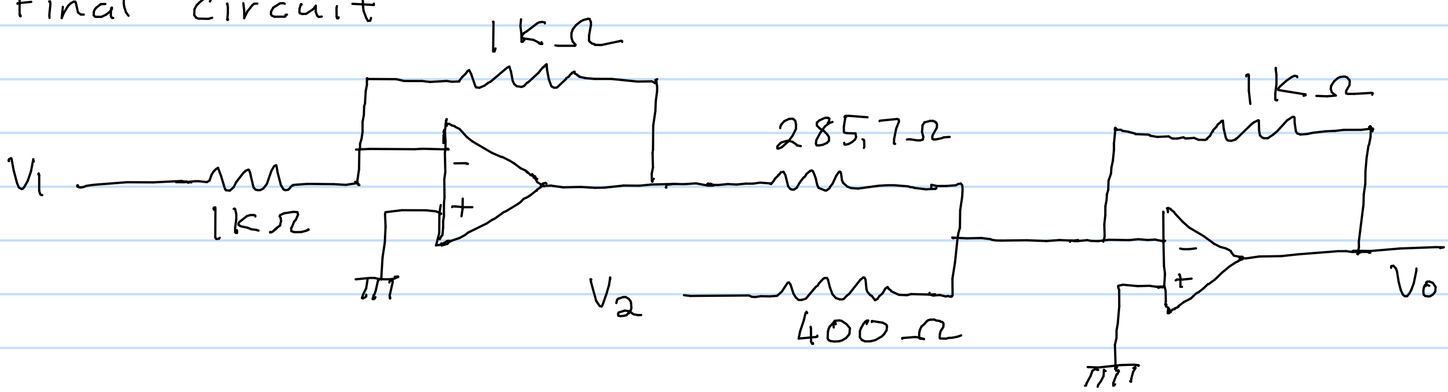
$$V_o = 3.5 V_1 - 2.5 V_2$$

$$= -(-3.5 V_1 + 2.5 V_2)$$

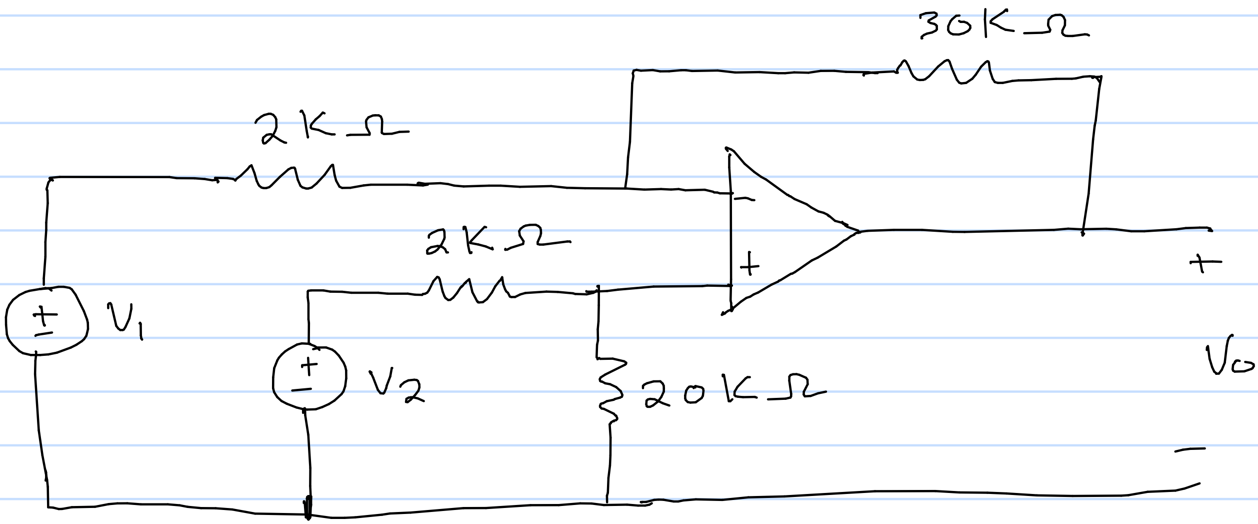
$$= -\left[\frac{3500}{1000} \cdot \left(\frac{-1}{1}\right) V_1 + \frac{2500}{1000} V_2\right]$$

$$= -\left[\frac{1000}{285.714} \left(\frac{-1000}{1000}\right) V_1 + \frac{1000}{400} V_2\right]$$

Final circuit



5.47 Find V_o, given $V_1 = 1\,V$ & $V_2 = 2\,V$



$$V_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} V_2 - \frac{R_2}{R_1} V_1$$

$$V_o = \frac{15(1 + 1/15)}{1 + 1/10} V_2 - 15 V_1$$

$$V_o = 14.545 V_2 - 15 V_1$$

$$= 14.545 \times 2 - 15$$

$$= 14.09\ V$$